# UTKARSH SHARMA

**Software Engineer & Systems Builder**

B.Tech in Computer Science & Engineering (Specialization: Data Science)

**GitHub:** github.com/shutkarshX
**LinkedIn:** in/utkarsh-sharma
**Production App:** tein.onrender.com

| 2,600+ | 1,830+ | 190+ | Docker / Cloud |
|---|---|---|---|
| TOTAL COMMITS | CONTRIBUTIONS | PASSING TESTS | LIVE PRODUCTION |

## 01 / PROFILE — OVERVIEW

Hands-on software engineer with proven experience architecting and deploying full-stack web platforms, native systems utilities in Rust, AI crisis command centers, and operating system test suites. Track record of rigorous testing discipline (over 190+ passing unit/E2E tests), clean system modularity, and live cloud delivery on Render. Focused on building high-reliability tools that solve concrete operational problems.

## 02 / CAPABILITIES — TECH STACK

### LANGUAGES

`Python` `TypeScript` `JavaScript` `Rust` `C (C99)` `Java` `SQL` `R`

### FRAMEWORKS & LIBRARIES

`FastAPI` `Flask` `React` `Vite` `Tailwind CSS` `Tauri 2` `Pandas` `Scikit-Learn`

### ARCHITECTURE & SYSTEMS

`PWA (Service Workers)` `Docker` `Win32 API` `REST APIs` `Asyncio` `TCP Sockets` `Leaflet GIS`

### DEVOPS & VERIFICATION

`Git / GitHub Actions` `Render Cloud` `Pytest (Automated)` `Chrome MV3` `Linux CLI` `Gunicorn`

## 03 / TRACK RECORD — VERIFIED PROJECTS

### TEIN — Attendance Intelligence & Campus Hub

`PRODUCTION` tein.onrender.com ↗ Showcase Repo ↗

Full-stack, mobile-first university platform deployed live on Render to solve attendance compliance and campus organization across semester cycles.

- **Bunk & Recovery Logic:** Implemented decision algorithms calculating exact safe lecture skip thresholds and mandatory recovery chains to safeguard the institutional 75% limit.
- **What-If Simulator:** Engineered a forward-looking planner allowing students to simulate hypothetical leave scenarios across future dates before taking time off.
- **Campus Community Hub:** Built an integrated directory indexing profiles, domains, and event broadcasts for 20+ university societies and technical clubs.
- **Offline PWA Architecture:** Implemented service worker caching (`/sw.js`) enabling fast, reliable operation over slow campus networks.

`Python 3` `Flask` `Docker` `PWA` `Gunicorn` `Render Cloud` `25 Unit Tests`

### RESQ-AI — Intelligent Disaster Response Command Center

github.com/shutkarshX/resq-ai ↗

Emergency operations command platform designed for municipal crisis management, real-time rescue prioritization, and geospatial team routing during flood disasters.

- **AI Incident Triage:** Integrated Google Gemini AI to parse citizen emergency dispatches, extract geospatial coordinates, and classify urgency severity.
- **Geospatial Command Map:** Rendered interactive Leaflet maps plotting active rescue convoys, evacuation shelters, and live incident clusters.
- **Async REST API:** High-throughput FastAPI backend with typed Pydantic request models and interactive OpenAPI/Swagger documentation.

`FastAPI` `React` `Gemini AI` `Leaflet Maps` `Python Async` `Swagger`

### Layer-Flow — Transparent Screen Drawing & Desktop Canvas

github.com/shutkarshX/Layer-Flow ↗

Native, lightweight Windows desktop application allowing presenters and educators to draw, highlight, and annotate directly over any active screen in real time.

- **Rust & Tauri 2 Architecture:** Sub-50MB memory footprint and zero UI latency, avoiding heavy Chromium wrapper overhead.
- **Win32 Hook Integration:** Implemented low-level OS hooks for global keyboard hotkeys and click-through pass-through state toggling.

`Rust` `Tauri 2` `React` `Win32 API` `Systems Programming`

### Speedo — Coursework Progression & Adaptive Video Controller

github.com/shutkarshX/Speedo ↗

Manifest V3 Chrome extension designed to automate learning progression, bypass player speed restrictions, and safeguard coursework progression.

- **CSP-Safe Injection:** Implemented a Main World execution bridge adhering to strict Manifest V3 Content Security Policies without security violations.
- **Modal Dismissal Guard:** Automatically intercepts and handles idle detection popups to maintain uninterrupted video progression.

`JavaScript` `Chrome Extension MV3` `DOM Automation` `CSP Safe`

### Integrated National Citizen Services & CPU Scheduling Lab

github.com/shutkarshX/...-Platform ↗

Systems engineering lab implementing and verifying fundamental process scheduling algorithms with an exhaustive automated test suite.

- **Core Scheduling Algorithms:** Implemented First-Come-First-Served (FCFS), Shortest Job First (SJF Preemptive & Non-Preemptive), Priority Scheduling, and Round Robin.
- **166 Automated Tests:** Rigorous test suite validating turnaround time, process starvation resistance, context switching, and CPU utilization.

`Python` `Operating Systems` `CPU Scheduling` `166 Automated Tests`

### Data Science & Predictive Modeling Suite (Prodigy Infotech)

github.com/shutkarshX/Prodigy-Infotech ↗

Data science suite covering exploratory data analysis, predictive decision tree modeling, sentiment mining, and geospatial accident analysis.

- **Decision Tree Classification:** Built and tuned decision tree classifiers on customer demographics to predict subscription behavior.
- **EDA & Sentiment Mining:** Analyzed demographic distributions, Titanic survival factors, and mined sentiment patterns from social media text data.

`Python` `Pandas` `Scikit-Learn` `Seaborn` `EDA`

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